

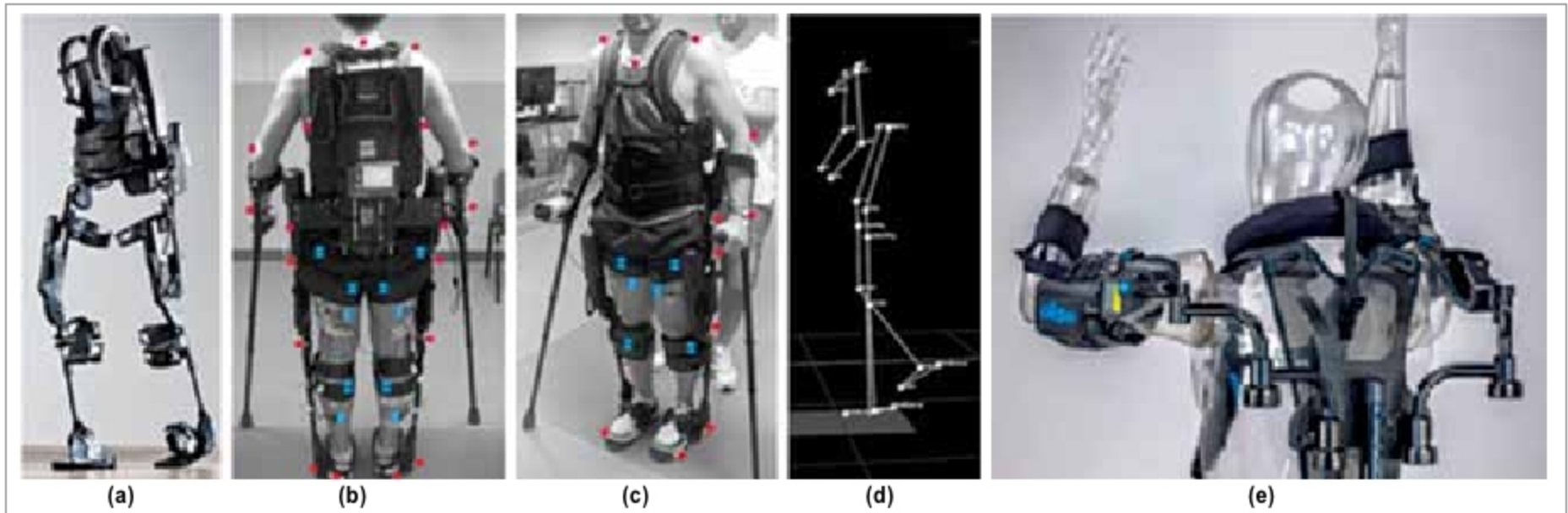


Fig. 5: (a) It is a battery-powered, wearable exoskeleton that enables individuals with severe walking impairments to stand up and walk over ground, (b, c) participants wearing the Ekso. Red dots indicate the positions of the reflective markers for the motion analysis and the blue rectangles indicate the position of the surface electromyographic (EMG) electrodes, (d) markers recorded by the motion analysis system (lateral view in the centre of the motion analysis lab pathway), and (e) exoskeleton for upper limbs [Credit: The Ekso (Ekso Bionics, Richmond, CA, USA)]

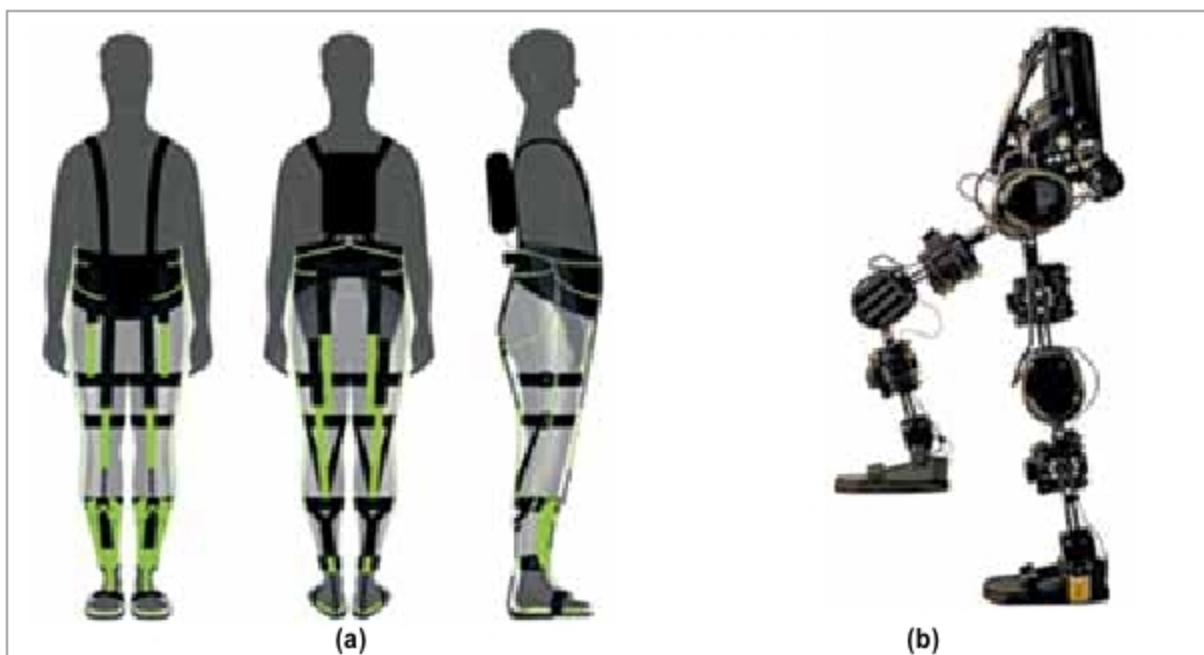


Fig. 6: (a) Exoskeleton for gait (Credit: XoSoft), (b) Exoskeleton for gait rehabilitation (Credit: Zarya)



Fig. 7: (a) Managing muscular dystrophy competently (Credit: GenElek), (b) Gait enhancing and motivating system (Credit: Samsung GEMS)

balance and coordination of hundreds of muscles, bones, joints, and a highly complex central nervous system. Gait disorders occur when the walking pattern is affected due to arthritis, inner ear disorder, stroke, foot conditions, neurological condition, and also due to the usage of poor-fitting shoes. Small motors attached in these devices

return movements to users' hips and knees. Refer pictures from XoSoft, GenElek and Samsung GEMS (Gait Enhancing and Motivating System).

Exoskeleton for paraplegics and post-stroke patients

A paraplegic person is capable of exercising, walking, and moving

easily with the help of exoskeletons. For stroke patients, there is a disrupt in the blood supply in brain, resulting in the death of brain cells. Loss of brain function leads to paralysis of one or more limbs. Ninety per cent of the stroke patients suffer from partial paralysis, leading to abnormal gait.

Exoskeletons for rehabilitation assist the joints to maintain balance while walking. These suits come with specially designed shoes. Users have to pull a cord attached to the shoe insole, and this provides the necessary torque to the ankle, thus correcting the abnormal walking motion.

Exoskeletons for cerebral palsy and spinal muscular atrophy in children

Spinal muscular atrophy in a child occurs when the nerves connecting the body and brain are disrupted, resulting in loss of control over the limbs and torso. Cerebral palsy in children occurs due to abnormal development or damage in the parts of the brain, leading to spinal deformation, delayed growth, and loss of motor muscle control.

Our brain has a specific part that controls our movement. When that part is damaged, the child cannot move his or her muscles normally.

Engineering behind exoskeletons

Design. Exoskeletons consist of rigid frame actuators that control the spherical motion of each hip and joints, microprocessors, software,